A Hybrid Flare-Gated Reduced Reference Model for Wormhole-Support Infrastructure

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Abstract

This paper presents a reduced reference model for hypothetical wormhole-support infrastructure under the conditional assumption that controlled null-energy-violating support can be supplied. The model is built from a spherically symmetric metric family

$$ds^2 = -N(l,t)^2 \mathrm{d}t^2 + B(l,t)^2 \mathrm{d}l^2 + R(l,t)^2 \mathrm{d}\Omega^2,$$

and uses a phase-structured lifecycle:

flattened R standby $\to$ B prestretch $\to$ R flare opening
$\to$ quiet access hold $\to$ R closure $\to$ hybrid repayment and shoulder matching
$\to$ B reset.

The construction separates the support plant into metric and source roles: $B(l,t)$ supplies radial proper-length support dilution, $R(l,t)$ supplies flare/access-state gating, $N(l,t)$ supplies timing and shoulder matching, and the source sector splits into nonminimally-coupled-scalar-like support, infrastructure repayment, and shoulder/matching components. The final reduced closure embeds the source ledger in an anisotropic stress tensor with conservation-completing radial flux. The flux-completed full-cycle ledgers close for the core, access, support, support-ring, and shoulder observer families, with a reduced conservation residual of approximately 5.7×10^{-4} relative to the source scale. A Lorentzian sampled-stress screen remains the active quantum-source gate. The paper therefore supplies a reproducible engineering reference model and a precise source-admissibility target for future quantum-field, field-equation, and backreaction analysis.

1 Purpose and contribution

Traversable wormhole physics supplies a strong constraint map. Morris and Thorne's throat construction makes flare-out and exotic support explicit [1]. Ford and Roman's quantum-inequality analysis ties negative energy to magnitude-duration restrictions and drives macroscopic wormhole support toward Planck-scale or large length-scale-discrepancy regimes [2]. Quantum interest turns repayment into a timed source-history problem [3]. Fewster and Roman's null-contracted sampling bound supplies a direct gate for wormhole stress histories sampled along timelike worldlines [6]. Dynamic throat work keeps local null-expansion and energy-condition diagnostics central [7].

The present paper contributes an engineering reference model inside that constraint map. It converts the phrase "wormhole exotic support" into a reduced infrastructure plant with separate roles, observer-family ledgers, source components, and closure checks. The contribution is a worked target model rather than a source-viability theorem. The model is useful because it identifies which part of a hypothetical support plant carries each burden and where the remaining physics gate acts.

The design sequence is:

$$\begin{aligned} &B\text{-only adiabatic stretch} \to B, R \text{ flare gating} \to B, R, N \text{ compensated reference geometry} \\ &\to \text{hybrid source architecture} \to \text{flux-completed reduced closure.} \end{aligned} \tag{1}$$

The end point is the Hybrid Flare-Gated Reduced Reference Model v1.0. It is a frozen reduced model for subsequent source-admissibility and backreaction work.

2 Literature setting

The model begins from the standard traversable-wormhole lesson: an areal-radius minimum with flare-out requires null-energy-condition violation near the throat [1]. Volume-integral work by Visser, Kar, and Dadhich showed that some wormhole geometries can make the integrated amount of energy-condition violation small, which motivates a separation between local peak burden and integrated burden [4, 5]. Fewster and Roman then showed how null-contracted quantum inequalities constrain such proposals and related slow-flare models when the exotic source is treated as a quantum field [6]. This sequence motivates the model's use of both local ledgers and sampled-stress ledgers.

The time-dependent side of the literature supplies the dynamic gate. Hochberg and Visser showed that dynamic wormhole throats retain local null-energy-condition pressure and require local throat definitions based on null congruences [7]. Kuhfittig's static and dynamic Ford–Roman-compatible constructions record the persistence of weak-energy-condition pressure in dynamic variants and the importance of traversal and redshift conditions [8]. These results motivate the model's controlled use of $R(l, t)$ as a flare-state gate and its explicit tracking of access exposure.

The source-candidate literature motivates the hybrid source architecture. Nonminimally coupled scalar fields can violate standard energy conditions, including ANEC, and support static spherically symmetric wormhole branches under positive curvature coupling [9]. In a different setting, Gao, Jafferis, and Wall demonstrated controlled traversability in an AdS/holographic construction through a double-trace deformation that generates negative averaged null energy and renders an Einstein–Rosen bridge traversable [10]. These source examples serve different roles here. The nonminimal scalar branch motivates the core/access support component. The holographic branch provides a benchmark for the broader idea that controlled couplings and source histories can generate traversability in special settings.

The paper's engineering language is also aligned with recent numerical exotic-spacetime screening practice, in which metric functions and source diagnostics are treated as design variables before source construction is attempted [11]. The present work specializes that design-variable approach to wormhole-support infrastructure.

3 Reduced metric family and observer roles

The reduced metric family is

$$ds^2 = -N(l, t)^2 \mathrm{d}t^2 + B(l, t)^2 \mathrm{d}l^2 + R(l, t)^2 \mathrm{d}\Omega^2. \tag{2}$$

The coordinate l labels the radial infrastructure coordinate. The metric controls have the following engineering assignments:

$$\begin{aligned} B(l,t) &\to \text{radial proper-length and support-dilution control,} \\ R(l,t) &\to \text{areal flare and access-state control,} \\ N(l,t) &\to \text{timing, redshift, shoulder matching, and isolation control.} \end{aligned}$$

The observer families used in the screens are the core line, access mean, support mean, support-ring mean, shoulder mean, and matching-region mean. Each family receives its own negative exposure, positive exposure, flux, and sampled-stress ledger.

The null-expansion prototype used for dynamic screens is

$$\theta_{\pm} = \frac{2}{R}\left(\frac{R_t}{N} \pm \frac{R_l}{B}\right). \tag{3}$$

The radial null-contracted source diagnostic is represented by $T_{kk}^{\pm}$ for the two radial null directions, with the stricter flux-completed readout

$$T_{kk,\min} = T_{kk} - 2|f|, \qquad f = T_{\hat{t}\hat{l}}. \tag{4}$$

This readout gives the unfavorable null direction when radial flux is present.

4 From adiabatic B control to flare-gated geometry

The initial adiabatic protocol used $B(l,t)$ as a radial infrastructure actuator. The purpose was to prepare a long proper-radial throat, hold a quiet access interval, and reset smoothly. The ramp rates, radial flux proxies, and acceleration proxies scaled cleanly under slower ramps. The source-history screen then assigned a persistent negative null-history ledger to the long setup and reset phases. This made B a strong support-dilution actuator and identified the need for a separate access-state control.

The next design step introduced $R(l,t)$ as the flare/access-state actuator. In the standby state, the near-core flare curvature is flattened. After B has been prestretched, R restores the flared access geometry for the access interval. After the access hold, R closes the flare before repayment and reset. The resulting lifecycle is

$$\begin{aligned} &\text{flattened } R \text{ standby} \to B \text{ prestretch} \to R \text{ flare opening} \\ &\to \text{quiet access hold} \to R \text{ closure} \to \text{repayment} \to B \text{ reset.} \end{aligned} \tag{5}$$

This order lets the support geometry be prepared while the access flare is closed, keeps the access hold quiet, and moves repayment outside the access interval.

5 Frozen reference geometry

Reference Geometry v0.3 is the frozen geometry used for the final reduced closure. Its parameters are given in Table 1.

The geometry screen shows strong access isolation during compensation. The access flux and $|K^{\theta}{}_{\theta}| \sim |R_t/(NR)|$ are concentrated in the intended flare-open and flare-close gates, while compensation occurs after flare closure. The invariant packet reports bounded N, bounded R, and controlled shoulder diagnostics. In the frozen geometry, B is effectively static during the open/access interval and carries the long-throat support-dilution role.

Table 1: Frozen geometry and access-compensation parameters for Reference Geometry v0.3.

Parameter	Value
B_0	8
w_B	10
T_B	150
T_R	5
T_H	60
T_C	20
Access compensation support amplitude	0.0082
Access compensation shoulder amplitude	0.0019
Shoulder center	2.3
Shoulder width	1.0
Shoulder lapse amplitude	-0.18

6 Hybrid source architecture

Source-realism screening converts the effective source into role-separated components. The final architecture is

$$T_{\mu\nu}^{\text{total}} = T_{\mu\nu}^{\text{NMC support}} + T_{\mu\nu}^{\text{infrastructure repayment}} + T_{\mu\nu}^{\text{shoulder/matching}}. \tag{6}$$

The nonminimally-coupled-scalar-like component carries the core/access support role during the open interval. The infrastructure repayment component carries support-ring and setup/reset repayment. The shoulder/matching component supplies buffer, matching, and shoulder repayment.

The component separation screen selected the hybrid architecture after the single-source scalar fit showed strong core/access matching and weak global matching. A locality-polished split gives the support component a clear role: it carries most of the core/access open burden while leaving support-ring and shoulder ledgers to infrastructure components. Table 2 summarizes the burden carried by the locality-polished NMC-like component.

Table 2: Open burden fraction carried by the locality-polished NMC-like support component.

Observer family	Open burden carried by NMC-like component
Core line	0.781
Access mean	0.924
Support mean	0.779
Support-ring mean	0.220
Shoulder mean	0.422

The strict-pass infrastructure amplitudes used in the final closure are

$$A_{\text{support,set/reset}} = 0.022, \qquad A_{\text{shoulder,set/reset}} = 0.016. \tag{7}$$

These values close the support-ring and shoulder ledgers after the flux-completed readout.

7 Reduced anisotropic closure

The source ledger is embedded in a reduced anisotropic tensor of the form

$$T^{\mu}{}_{\nu} = \mathrm{diag}(-\rho, p_r, p_t, p_t) + T^{t}{}_{l}. \tag{8}$$

For each null-contracted source target, the reduced closure uses

$$\rho = \frac{1}{2}T_{kk}, \qquad p_r = \frac{1}{2}T_{kk}, \qquad p_t = 0. \tag{9}$$

The conservation-completing radial flux $f = T_{\hat{t}\hat{l}}$ is computed from the reduced continuity condition

$$\partial_t \rho + \frac{1}{BR^2}\partial_l(NR^2 f) \approx 0. \tag{10}$$

The final model is therefore an actuator-exchange closure. The closure supplies a bounded radial transport/exchange channel for the infrastructure source and leaves a small residual after flux completion.

The global residual is

$$\frac{\max|\mathrm{residual}|}{\max|T_{kk}|} = 5.701 \times 10^{-4}. \tag{11}$$

The required radial flux scale relative to the source scale is listed in Table 3.

Table 3: Required conservation-completing radial flux relative to the source scale.

Zone	Required flux/source scale
Access	0.132
Support	0.138
Shoulder	0.097

8 Full-cycle flux-completed ledgers

The strict readout applies Eq. (4) to account for the unfavorable radial null direction. The full-cycle positive-to-negative ratios are given in Table 4.

Table 4: Flux-completed full-cycle positive-to-negative ratios for the primary observer families.

Observer family	Flux-completed full-cycle ratio
Core line	1.083
Access mean	1.471
Support mean	2.801
Support-ring mean	1.135
Shoulder mean	1.130

These results are the engineering closure of the reduced model. The source architecture, radial flux channel, and repayment phases produce closed ledgers for the core, access, support, support-ring, and shoulder families.

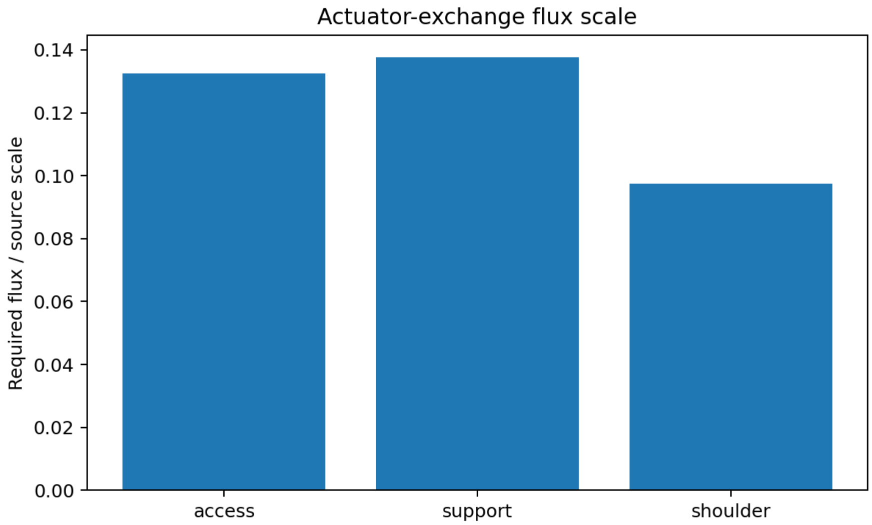

Figure 1: Actuator-exchange flux scale by zone. The reduced closure keeps the flux-completing channel bounded relative to the source scale.

9 Sampled-stress physics gate

The active quantum-source gate is the Lorentzian sampled-stress proxy. The screen uses

$$\langle T_{kk}\rangle_\tau \gtrsim -\frac{C}{\tau^4}, \qquad C = \frac{3}{32\pi^2}. \tag{12}$$

The flux-completed source has negative margins for the sampled windows listed in Table 5. The table gives the most constraining window from the reduced screen for each observer family.

Table 5: Worst Lorentzian sampled-stress proxy margins for the flux-completed source.

Observer family	Worst margin	Window τ	Center
Core line	-1.062×10^{-3}	5.0	177.06
Access mean	-1.032×10^{-3}	5.0	177.06
Support mean	-8.215×10^{-3}	2.0	8.58
Support-ring mean	-2.746×10^{-2}	2.0	8.58
Shoulder mean	-8.607×10^{-3}	2.0	8.58

This screen gives the next target precisely. The reduced engineering model closes geometry and ledgers, and the source-admissibility phase now evaluates the frozen source target against source-specific quantum inequalities, quantum-interest timing, NMC scalar field equations, and backreaction.

10 Engineering interpretation

The model supplies a complete reduced engineering architecture for the current phase. The geometry control, observer ledgers, source roles, and closure check are connected by a single workflow:

$$\text{geometry control} \to \text{observer-family ledger} \to \text{source role} \to \text{closure check} \to \text{physics gate}. \tag{13}$$

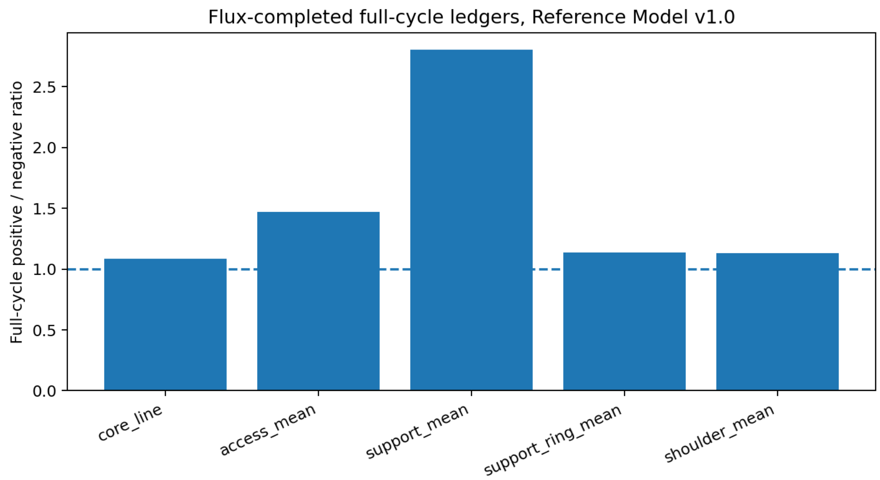

Figure 2: Flux-completed full-cycle ratios. All primary observer-family ledgers clear unity in the reduced closure.

The main design-level findings are:

1. $B(l,t)$ is a support-dilution actuator. It reduces local support severity and prepares the radial infrastructure.
2. $R(l,t)$ is a flare/access-state actuator. It opens the flared access state after prestretch and closes it before repayment.
3. $N(l,t)$ is a timing and matching actuator. It shapes the shoulder and supports access isolation.
4. The source sector is best organized by roles. The NMC-like component carries core/access support, while infrastructure and shoulder components carry repayment and matching.
5. Flux-completed anisotropic closure converts the source ledger into a reduced actuator-exchange tensor with bounded flux and small residual.
6. The Lorentzian sampled-stress screen supplies the active quantum-source gate and turns the remaining physics problem into a concrete target.

This organization is the paper's scientific and engineering utility. It makes the source problem reproducible and decomposed.

11 Reproducibility package

The accompanying bundle contains the LaTeX source, PDF, scripts, data tables, figures, literature access notes, and manifest. The main scripts are:

- `run_invariant_packet_v03.py`: frozen-geometry invariant and observer-family diagnostics;
- `run_source_realism_prescreen_v02.py`: phase-structured source ledger screen;
- `run_candidate_source_model_v02.py`: full NMC scalar-channel compatibility screen;
- `run_candidate_source_model_v03_component_polish.py`: component separation and locality polish;

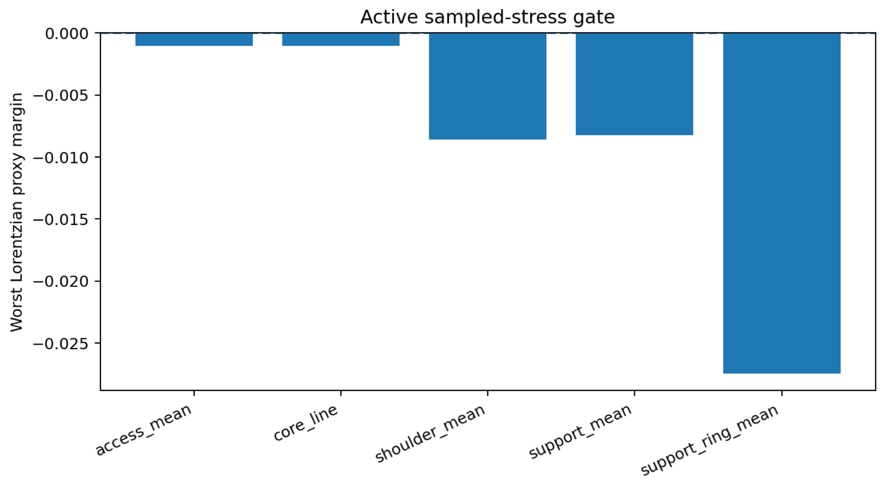

Figure 3: Worst Lorentzian proxy margins for the frozen reduced model. The sampled-stress result supplies the active physics gate for the next source-admissibility phase.

- `run_candidate_source_model_v03p1_residual_balance.py`: residual infrastructure-repayment balance;
- `run_hybrid_source_closure_v01.py`: reduced anisotropic closure;
- `run_hybrid_source_closure_v01p1_fluxmin_rebalance.py`: strict flux-completed rebalance.

The data folder includes the observer ledgers, closure summaries, sampled-stress tables, source-component scores, and extracted summaries used in the paper.

12 Scope and next phase

The frozen result is

$$\boxed{\text{Hybrid Flare-Gated Reduced Reference Model v1.0}}. \tag{14}$$

Its status is summarized in Table 6.

The next phase evaluates quantum-source admissibility for the frozen target. The work items are: field-equation closure for the NMC-like support component, actuator-exchange interpretation for infrastructure repayment, source-specific quantum inequalities, quantum-interest timing for each negative/positive source pair, perturbative backreaction across flare opening and closure, and repeated-operation accumulation in the support-ring and shoulder components.

13 Conclusion

This paper constructs a phase-structured reduced reference model for wormhole-support infrastructure. The model uses B as a radial support-dilution actuator, R as a flare/access-state actuator, N as a timing and shoulder-matching actuator, and a hybrid source sector that separates NMC-like support from infrastructure repayment and shoulder/matching. The final reduced closure embeds the source ledger in an anisotropic tensor with conservation-completing radial flux. The flux-completed full-cycle ledgers close for the core, access, support, support-ring, and shoulder observer families, and the conservation residual remains small in the reduced embedding.

Table 6: Gate status for the reduced reference model.

Gate	Status
Geometry lifecycle	Frozen as Reference Geometry v0.3
Source architecture	Frozen as Hybrid Source Architecture v0.3
Component separation	NMC support plus infrastructure repayment plus shoulder/matching
Flux-completed ledgers	Primary observer-family ratios above unity
Reduced conservation	Small residual with bounded actuator-exchange flux
Access isolation	Preserved in the reduced screens
Lorentzian sampled-stress proxy	Active quantum-source gate
Physical source construction	Next phase
Backreaction/stability	Next phase

The model supplies a frozen target for the next physical gate. Its Lorentzian sampled-stress screen identifies the active quantum-source admissibility problem with explicit observer families, sampling windows, and margins. The result is therefore a reproducible engineering reference model and a concrete source-admissibility benchmark for future semiclassical, scalar-field, and backreaction studies.

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